Tor Browser Documentation

A guide to using the browser to reach the dark web safely

Technical documentation · British English
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1. Introduction

Tor Browser is a free, open-source browser maintained by the Tor Project and designed to protect the privacy and anonymity of those who browse with it. It is based on Firefox ESR and sends traffic through the Tor network, which is run by volunteers around the world. This documentation explains how Tor Browser works, how to install and configure it, how to reach onion services (the so-called dark web) and what precautions to take.

The content is informative and aimed at legitimate use of the tool: protecting privacy, circumventing censorship and safely accessing legitimate onion services. This document does not list or recommend websites and does not cover access to illegal content.

Updates: Tor Browser follows the Firefox release cycle, which became shorter in 2026. Menu names may vary between versions; if in doubt, consult the official manual, available in the browser itself (Help > Tor Browser Manual) and on the Tor Project website.

2. What Is the Dark Web?

The internet is often described in layers. The **Surface Web** is the part indexed by search engines. The **deep web** covers everything that is not indexed, such as email, online banking, intranets and databases, and it is vastly larger than the surface. The **dark web** is a small part of the deep web that can only be reached with specific software, such as Tor, and whose addresses end in **.onion**.

- The deep web and the dark web are **not synonyms**: most of the deep web is ordinary and lawful, like your email inbox.
- The dark web hosts legitimate services (news outlets, whistleblowing platforms, email services and communities that need protection) as well as criminal activity.
- Simply using Tor does not make a person suspicious, but what is done on it remains subject to the law.

3. How Tor Works

Instead of connecting directly to a website, Tor routes traffic through a series of three volunteer servers, called **relays**, forming a circuit. The data is encrypted in layers, like those of an onion, and each relay removes only one layer.

Stage	Role	What it knows
Entry relay (guard)	Receives the user's connection	Your IP address, but not the destination
Middle relay	Passes the traffic along	Only the previous and the next relay
Exit relay	Delivers the traffic to the destination site	The destination, but not the user's IP address

As a result, no single relay knows both who you are and what you are accessing. The circuit is renewed periodically, and the browser may use different circuits for different sites.

Onion services

With onion services, traffic never leaves the Tor network: user and server meet at a rendezvous point inside it, and neither reveals its IP address to the other. Current (version 3) addresses are 56 characters long followed by **.onion** and are derived from a cryptographic key, which makes them long and hard to memorise.

Note: because the traffic of an onion service never passes through an exit relay, it remains protected by encryption along the entire route inside the network.

4. Legitimate Uses

- **Journalists and sources**: protected communication and whistleblowing platforms such as SecureDrop.
- **Activists and human rights defenders**: protection against surveillance and reprisals.
- **People in countries with censorship**: access to blocked sites and services.
- **Ordinary users**: less tracking by advertisers, providers and public networks.
- **Researchers and security professionals**: study of the network and of threats, within the law.
- **Services with an onion version**: some news outlets and online services maintain onion addresses to offer safer access.

5. Installation

1. Go to the official Tor Project website (torproject.org) by typing the address directly into your browser.
2. Download Tor Browser for your system: Windows, macOS, Linux or Android. There is no official Tor Project version for iOS; the project recommends the Onion Browser app for iPhone and iPad.
3. Optional but recommended: verify the digital signature of the downloaded file, following the instructions on the official website.
4. Install and open the program. On first launch, click **Connect**.
5. Wait for the circuit to be established; once it is, the browser will open its home page.

Warning: download Tor Browser only from the official website. Tampered versions, distributed by fake sites, may contain malicious software that compromises the very anonymity you are trying to protect.

6. First Connection and Bridges

In some countries, networks or providers, direct connection to the Tor network is blocked. In such cases a **bridge** is used: a relay that is not publicly listed, which makes blocking harder.

- **Connect**: the default option, used when there is no blocking.
- **Configure Connection**: lets you choose bridges or set up a proxy.

- **Snowflake**: a type of bridge that uses temporary connections provided by volunteers and is suited to getting around blocks.
- **obfs4**: a type of bridge that disguises Tor traffic so that it looks ordinary.
- **Connection Assist**: a browser feature that helps detect blocking and suggest bridges.

7. Interface and Security Levels

- **Onion icon and circuit information**: show which countries the current tab's connection passes through and let you request a new circuit for the site.
- **New Identity**: closes all tabs, clears the session's history and cookies and uses new circuits, as if it were a fresh session.
- **NoScript**: a built-in extension that controls the running of scripts according to the security level.
- **Standard window size**: Tor Browser adjusts content to standardised sizes (letterboxing) to make identification by screen size more difficult.

Security levels

Accessed through the shield icon or under **Settings > Privacy and security**. In general terms:

Level	Behaviour	Suited to
Standard	All website features enabled	General use on trusted sites
Safer	Restricts higher-risk features, such as scripts on non-HTTPS sites, and requires a click to play media	Everyday use with extra caution
Safest	Disables scripts on most sites and limits fonts, images and other features; some sites may break	High-risk situations

Note: higher levels reduce the attack surface but leave many sites less functional. Choose according to the real risk of your situation.

8. Accessing Onion Services

1. Obtain the .onion address from a trustworthy source, ideally the organisation's official website on the Surface Web.
2. Type or paste the address into the Tor Browser address bar and check every character.
3. Some sites announce their onion address through a special header, and the browser shows a **.onion available** option.
4. Some services require key-based authentication; the browser will ask for the key when needed.

Beware of fake addresses: because onion addresses are long, scammers create sites with very similar addresses to steal accounts and money. Never trust anonymous lists of links, and confirm the address from more than one reliable source.

9. Good Practice

- Keep Tor Browser up to date at all times.
- Do not install additional extensions, as they can make your browser identifiable.
- Avoid maximising the window and do not change settings that alter your fingerprint.
- Do not use Tor to download large files by torrent: it is slow, harms the network and may expose your real IP address.
- Do not log in to accounts that identify you if your aim is to remain anonymous.
- Prefer HTTPS sites: Tor protects the route, but the exit relay can see data that is not encrypted.
- Do not open downloaded documents (such as PDFs and office files) while connected; they may make requests outside Tor.
- Use **New Identity** when you change activity.
- Do not combine Tor with other network tools without understanding the effects.

10. Limitations and Risks

- **It is not total invisibility**: Tor reduces tracking, but user error, malware or vulnerabilities can still expose an identity.
- **Lower speed**: the route through several relays makes browsing slower.
- **Blocks and CAPTCHAs**: many sites restrict or hinder access by Tor users.
- **Scams and phishing**: common among onion services, where there is no reliable identification of operators.
- **Disturbing or illegal content**: may be encountered by following unknown links.
- **Malware**: files downloaded from unknown sources can be dangerous.

Legal notice: using Tor is lawful in most countries, but some jurisdictions restrict it. Illegal activity remains illegal on the Tor network, and the authorities have investigative methods that do not depend on breaking the network's anonymity. Everyone is responsible for knowing and complying with the law of the country they are in.

11. Troubleshooting

Problem	What to do
Cannot connect to the Tor network	Check the system date and time, temporarily disable proxies and restrictive firewalls, and try using a bridge such as Snowflake.
Connection is very slow	Use **New Identity** or request a new circuit for the site; avoid heavy features such as video.
Site will not open or asks for a CAPTCHA	Try a new circuit; some sites block Tor, and it may be necessary to use the service's onion version, where one exists.
Site broken or missing images	The security level may be too high; lower it only if the site is trustworthy.
Invalid onion address warning	Re-check every character of the address and confirm it against the official source.

12. Conclusion

Tor Browser is an important tool for protecting privacy and freedom of information, especially for those who face surveillance or censorship. Using it effectively depends on installing the software from official sources, keeping it up to date and adopting careful habits. It is not a magic solution: anonymity depends as much on the user's behaviour as on the technology, and respect for the law remains everyone's own responsibility.